

I help companies transform ideas into successful products.

WORK EXPERIENCE

Google - Zurich

February 2022 - April 2026

AI/ML Engineer - Google Search Shopping

- Developed a **RAG** system to generate **LLM**-based product summaries for shopping queries in [AI Overviews](#), streamlining user navigation and boosting engagement.
- Trained and launched a **LoRA adapter** for shopping products, enabling efficient fine-tuning of the **LLM** to deliver faster, more relevant product summaries while reducing **TPU** consumption by 48%.
- Engineered a **LoRA-based** selection layer to identify representative products from candidate sets, successfully mimicking the performance of large-scale models while meeting strict latency requirements for real-time [AI Overviews](#).

Software Engineer - YouTube ads

- Pioneered the implementation, validation, and successful launch of [Target Frequency](#), a game-changing feature empowering advertisers to set weekly ad exposure targets. This initiative resulted in a remarkable revenue surge, surpassing \$200M in its inaugural year in 2023 and \$0.5B in 2024.
- Innovated by introducing the [Reserve Target Frequency](#) product in YouTube. This new offering seamlessly integrates premium reserve benefits with a frequency guarantee, revolutionizing user engagement, and generating \$15M in its first year.
- Effectively leveraged **C++**, **Python**, **SQL**, and internal frameworks to power these impactful innovations.

Software Engineer

UiPath - Bucharest

July 2020 – February 2022

- Elevated to the role of lead for the [system activities package](#), where I spearheaded the creation of new activities, maintained existing ones, and resolved bugs, providing effective guidance to a team of three. Leveraging technologies such as **C#** and **WPF**, we ensured the seamless functioning of critical system components.
- Engineered rule checkers for the Studio product, introducing a sophisticated mechanism to analyze user workflows. This innovation effectively identified errors and warnings in code, resulting in a significant reduction in post-publishing bugs.
- Pioneered the conceptualization and implementation of impactful features, driving substantial efficiency gains. These enhancements not only streamlined operations but also translated into time savings of over 1000 hours per month for each customer, underscoring a commitment to delivering substantial value.

Technical Engineering Intern

Microsoft - Bucharest

April 2020 – June 2020

- Expertly tackled challenges within Microsoft Teams stemming from a surge in new users during the pandemic lockdown, swiftly resolving issues to ensure seamless functionality.
- Empowered clients by swiftly addressing and remedying application-related issues, ultimately saving them valuable time and facilitating uninterrupted business operations.

Software Engineer

Amiq Consulting - Bucharest

July 2019 – April 2020

- Led the design and implementation of comprehensive analytics dashboards for regressions, providing crucial statistical insights into the development status of the internal team. Leveraging **Python**, **Elasticsearch**, and **Kibana**, I initiated this project from the ground up, ensuring accurate and insightful data representation.
- Innovatively crafted scenarios in **M-SDL** (Measurable Scenario Description Language) to rigorously test and assess the behavior of autonomous vehicles.

Part-time Software Engineer

4PSA - Bucharest

October 2018 – January 2019

- Engineered an efficient solution for legacy **C** functions, seamlessly transitioning from traditional file-based operations to streamlined streams input/output. This innovative approach significantly slashed the overhead associated with inter-process communication, enhancing efficiency and modernizing the functionality of the system.

Freelance Web and App Developer

Bucharest

October 2013 – September 2016

- Played a key role in developing new features (windows and web application) in **C#** for a loan management software company based in the USA.
- Features include dashboards for users to see the status of the loans, payment dialogs integrated in application, etc.

EDUCATION

Bucharest, Romania

Polytechnic University of Bucharest - Masters

October 2019 – June 2021

Artificial Intelligence M.Sc.

- Dissertation project: Video segmentation using convolutional neural networks.
- Undergraduate Modules: **Computer Vision, CUDA, NLP, Data Mining, Knowledge Representation and Reasoning, Robotics, Multi-agent system, Advanced Operating Systems, Artificial Intelligence, Neural Networks, Data Mining, Data Science, Symbolic and Statistical Learning, Self-Organizing Systems, Neural Networks, Advanced AI Techniques, Advanced Java programming**, etc.

Bucharest, Romania

Polytechnic University of Bucharest - Bachelor

October 2015 – June 2019

- Computer Science and Engineering Department, Artificial Intelligence.
- Final Thesis: Image segmentation using convolutional neural networks (Car segmentation)
- Undergraduate Modules: **Machine Learning, Artificial Intelligence, Algorithms Design, Algorithm Analysis, Programming Paradigms, Probability Theory, Data Structures, Databases (SQL and NoSQL), Parallel and Distributed Algorithms, Operating System, Compilers, Computer Graphics, Computer Systems' Architecture, Digital electronics, Computers Structure and Assembly Languages, Analog Electronics, Communication protocols, Electrical Engineering, Numerical Methods, Microprocessor Based System Design, Foundations of Security for Enterprises, Graphical Processing Systems, Human-Computer Interaction, Local Area Networks, Formal Languages and Automata, CAD-CASE Systems** etc.

OTHER TECHNICAL EXPERIENCE

Extracurricular Activities

- **Google HashCode** (2017 – **top 50%**, 2018 – **top 50%**, 2019 – **top 20%**)
- **Bestem Hackathon** (2018) - **ranked 1st** out of 20+ teams
 - Implemented in C# in a team of 3 a **trojan virus** that can control the victim's **command prompt**, copy and create **files**, **hide itself** in the Task Manager, **autorun** when the system is turned on. The virus also includes a **key logger**.
- **Microsoft Techathon** (2017) - **ranked 2nd** out of 15+ teams
 - Implemented applications in a team of 2 using Microsoft's **Computer Vision REST API** to transform handwritten text into digital text and analyze pictures to tag people's emotions.

Projects

- [Chatbot helping with order placements](#) (2020 NLP project)
 - Created a network using **BERT** which enables clients to create dinner reservations talking to a chatbot.
- **Semantic Segmentation - Depth Relationship** (2020 Computer Vision project)
 - I used encoder-decoder **CNN** models to prove the relationship between depth estimation and semantic segmentation estimation.
- [Image segmentation](#) (2019 bachelor thesis Computer Vision) – **10/10**.
 - I trained a **convolutional neural network (ResNet50)** pretrained on **ImageNet** with **SegNet** architecture) to segment cars in pictures with **97.34%** accuracy on the **test set**.
 - The network uses **dice loss** function, **adadelata** optimizer and **intersection over union** accuracy metric.
- [Genetic algorithm](#) (2019) – **10/10**.
 - Implemented algorithm that generates images using basic shapes (**ellipses** or **rectangles**) to match input pictures.
 - Using **roulette wheel selection**, I created images, with as little as 50 shapes, that resemble the original pictures.
- **Shell** (2018).
 - Implemented in **C** a shell which can run commands with arguments (external or internal), pipes and redirections.
- **Support-vector machine** (2019) – **10/10**.
 - Implemented classifiers using SVMs for both **binary** classification (skin or not skin dataset) and **multi-class** (type of news dataset). Kernels: **linear**, **polynomial**, **Gaussian** and **Sigmoid**. Evaluation: **K-fold cross validation**.

Other Languages and Technologies

- Keras, Java, C, C#, SQL, , TCP/IP stack & protocols, Tensorflow, PyTorch, CUDA, React, HTML & CSS, JavaScript, MATLAB, Haskell, Prolog, .Net MVC, Razor, Bash, Visual Studio, Eclipse, Octave, VirtualBox, SQL Server Management Studio.